Project Executable Files

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes — app.py, src/ modules, models/, data/
2	README / Setup Guide (instructions to run the project)	No — not currently in the repository; recommended addition before final submission
3	requirements.txt / package.json / dependency file	Yes — requirements.txt present
4	Database schema / seed files (if applicable)	NA — project uses CSV logging, not a database
5	Environment configuration file (.env.example or similar)	Partial — a working .env exists locally at the project root (holding GEMINI_API_KEY) and is correctly excluded from Git via .gitignore, but no .env.example template is committed for other developers to copy
6	Deployed application URL (if hosted)	No — application currently runs locally only
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	No — not containerised
9	Test files and test results	No — no automated test suite; testing was manual (see Performance Testing)
10	Demo video or walkthrough (if required)	[Add link before submission]

Step 2: File / Folder Structure

Project folder structure exactly as shown in the VS Code Explorer for this repository:

```
EMOTION-DETECTION/
  data/   — raw / reference datasets
  models/
    bert_emotion_model_final/
      bert_emotion_model_final.zip
      bert_label_encoder.pkl
      config.json
      model.safetensors
      tokenizer_config.json
      tokenizer.json
      training_args.bin
    bilstm/
      bilstm.keras
      label_encoder.pkl
      tokenizer.pkl
    bert_label_encoder.pkl   — duplicate copy kept at models/ root
  src/
    __pycache__/
    __init__.py
    bert_predict.py
    bilstm_predict.py
    gemini_helper.py
    keyword_enhancer.py
    logger.py
    mixed_emotion.py
    prediction_schema.py
    preprocessing.py
    utils.py
  venv/   — local virtual environment (not committed)
  .env   — local Gemini API key (excluded from Git via .gitignore)
  .gitignore
  app.py   — Streamlit application entry point
  emotion_response_examples.csv   — interaction log / example data
  requirements.txt
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	[Not yet deployed — add hosted URL here if deployed, e.g. Streamlit Community Cloud]
Login Credentials (Demo)	Not applicable — the app has no authentication / login
Platform / Hosting Provider	[e.g. Streamlit Community Cloud, if deployed]
Repository Link	[Add GitHub repository link here before submission]
Demo Video Link	[Add link before submission]

Step 4: Run Instructions

- Clone the repository and open a terminal in the project root.
- Create and activate a virtual environment (kept local to the project as venv/, already present and excluded from Git), then run: pip install -r requirements.txt
- Create a .env file in the project root containing: GEMINI_API_KEY=your_api_key_here (a working .env already exists locally for development; it is not committed, per .gitignore).
- Run the app with: streamlit run app.py

- Open the local URL Streamlit prints in the terminal (default http://localhost:8501) and start describing a learning problem.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	app.py imports predict_bilstm and EMOTION_RESPONSES twice (redundant duplicate import lines).	Cosmetic only — does not affect functionality; recommended cleanup.
2	Sidebar 'CSV Examples' counter reads emotion_responses.csv, but src/logger.py actually writes to emotion_response_examples.csv, so the sidebar count never reflects real logged rows.	Open — recommend aligning both filenames to the same constant.
3	bert_label_encoder.pkl exists in two places (models/bert_emotion_model_final/ and models/ root), which is easy to load from the wrong path by mistake.	Open — recommend keeping a single copy and updating any hard-coded paths.
4	No persistent, multi-session history — emotion_history lives only in Streamlit session state and resets on refresh.	Open — could be moved to the CSV log or a lightweight database.
5	Broad except Exception blocks around the Gemini call and predictions can mask the specific cause of a failure.	Open — recommend narrowing exception handling and logging errors.
6	No automated tests currently exist for the prediction pipeline.	Open — recommend adding unit tests for preprocess_text, predict_bilstm and predict_bert.